

Cell populations at the start of organogenesis

CHAPTER 12

SPECIFICATION OF THE BODY AXES AND THE BODY PLAN

Embryos may be thought of as being constructed with three orthogonal spatial axes (cephalocaudal, dorsoventral and laterolateral), plus a temporal axis. In mammalian embryos, axes cannot be specified at very early stages; embryonic axes can be defined only after the early extra-embryonic structures have been formed and the inner cell mass can be seen. The position of the future epiblast can be predicted in human embryos when the hollow blastocyst has formed. The inner cell mass becomes (seemingly) randomly located on the inside of the trophoctoderm and forms a population of epiblast cells subjacent to the trophoblast. This region implants first. It is not known whether the trophoctoderm in contact with the inner cell mass initiates implantation, so that the future dorsal surface of the embryo is closest to the disrupted maternal vessels at the implantation site, or whether the inner cell mass can travel around the inside of the trophoblast to gain a position subjacent to the implantation site once implantation has started.

Axes may be conferred on the whole embryonic disc, which is initially flat and mainly two-dimensional. However, their subsequent orientation in the folded three-dimensional embryo will be completely different. The dorsal structures of the folded embryo form from a circumscribed central ellipse of the early flat embryonic disc (see Fig. 10.5). Lateral and ventral structures form from the remainder of the disc, and the peripheral edge of the disc eventually becomes constricted at the umbilicus (see Figs 10.9–10.10). Although the appearance of part of the epiblast is taken to specify the dorsal surface of the embryo, the inner layer, i.e. the hypoblast, is not, by default, a ventral embryonic structure.

The primary, cephalocaudal, axis is conferred by the appearance of the primitive streak in the bilaminar disc. The primitive streak patterns cells during ingression, and so also specifies the dorsoventral axis, which becomes apparent after embryonic folding. The position of ingression through the streak confers axial, medial or lateral characteristics on the forming mesenchyme cells. The axial and medial populations remain as dorsal structures in the folded embryo, and the surface ectoderm above them will exhibit dorsal characteristics. The lateral plate mesenchyme will assume lateral and ventral positions after embryonic folding, and the surface ectoderm above this population will gain ventral characteristics.

The third and last spatial axis is the bilateral, or laterolateral, axis, which appears as a consequence of the development of the former two axes. Initially, the right and left halves of the embryonic body are bilaterally symmetric. Lateral projections, the upper and lower limbs, develop in two places on each side of the body wall (somatopleure).

With the last axis established, the temporal modification of the original embryonic axes can be seen. The segmental arrangement of the cephalocaudal axis is very obvious in the early embryo and is retained in many structures in adult life. Similarly, dorsal embryonic structures remain dorsal and undergo relatively little change. However, structures that were originally midline and ventral, especially those derived from splanchnopleuric mesenchyme, e.g. the cardiovascular system and the gut, are subject to extensive shifts, and change from a bilaterally symmetric arrangement to an entire body that is now chiral, i.e. has distinct left and right sides.

The development of all the body organs and systems, organogenesis, begins after the dramatic events of gastrulation, when the embryo has attained a recognizable body plan. In human embryos, this corresponds to the end of stage 10 (Fig. 12.1). The head and tail folds are well formed, with enclosure of the foregut and hindgut (proenteron and metenteron), although the midgut (mesenteron) is only partly constricted from the yolk sac. The forebrain projection dominates the cranial end of the embryo, and the buccopharyngeal membrane and cardiac prominence are caudal and ventral to it. The cardiac prominence

contains the transmedian pericardial cavity, which communicates dorsocaudally with right and left pericardioperitoneal canals. These pass dorsally to the transverse septum mesenchyme and open caudally into the extraembryonic coelom on each side of the midgut. The intra-embryonic mesenchyme has begun to differentiate and the paraxial mesenchyme is being segmented into somites. Neural groove closure is progressing caudally, so that a neural tube is forming between the newly segmenting somites. Rostrally, the early brain regions, which have not yet fused, can be discerned. The neuroepithelium is separated from the dorsal aspect of the gut by the notochord. The earliest blood vessels have appeared and a primitive tubular heart occupies the pericardium. The chorionic circulation is soon to be established, after which the embryo rapidly becomes completely dependent on the maternal blood stream for its requirements. The embryo is connected to the developing placenta by a mesenchymal connecting stalk in which the umbilical vessels develop, and which also contains the allantois, a hindgut diverticulum. The lateral body walls are still widely separated. The embryo has contact with three different vesicles: the amnion, which is in contact with the surface ectoderm; the yolk sac, which is in contact with the endoderm; and the chorionic cavity, containing the extraembryonic coelom, which is in contact with the intraembryonic coelomic lining (see Fig. 10.10).

The early body plan of the embryo is segmented. The boundaries between the segments are maintained by the differential expression of genes and proteins that restrict cell migration in these regions. Organogenetic processes either retain the segmental plan, e.g. spinal nerves, or replace it locally, e.g. the modifications of somatic intersegmental vessels by the development of longitudinal anastomoses. Abnormalities may result from improper specification of segments along the cephalocaudal axis and may fail to produce the appropriately modified segmental plan.

The degree to which vertebrate embryos are developmentally constrained at this period of development is controversial. Comparative studies on the timing at which specific embryonic structures appear, heterochrony, have shown that other embryonic species do not follow the same developmental sequence as humans (Richardson and Keuck 2002). Although some developmental mechanisms are highly conserved, e.g. the homeobox gene codes, others may have been dissociated and modified in different vertebrate species during evolution.

Organogenesis, the further development of body regions and organs that is described elsewhere in this book, starts from about stage 10 (approximately 28 days). Although it is both conventional and convenient to consider the further development of each body system on an individual basis, not only do all systems develop simultaneously, but also they interact and modify each other as they develop. This necessary interdependence is supported by the evidence of experimental embryology and reinforced by the phenomena of growth anomalies, which cut across the artificial boundaries of systems in most instances. For these reasons, it is recommended that the development of an individual system or body region should be studied in relation to others, especially those most closely associated with it, whether spatiotemporally or causally.

EMBRYONIC CELL POPULATIONS AT THE START OF ORGANOGENESIS

The developmental processes operating in the embryo between stages 5 and 9 enabled the construction of the bi- and trilaminar embryonic disc, the intraembryonic coelom and new proliferative epithelia. From the end of stage 10, a range of local epithelial and mesenchymal populations now interact to produce viscera and appendages. The inductive influences on these tissues and their repertoire of responses are very different from those seen at the onset of gastrulation. The range of tissues present at the start of organogenesis, when the body plan is clear,

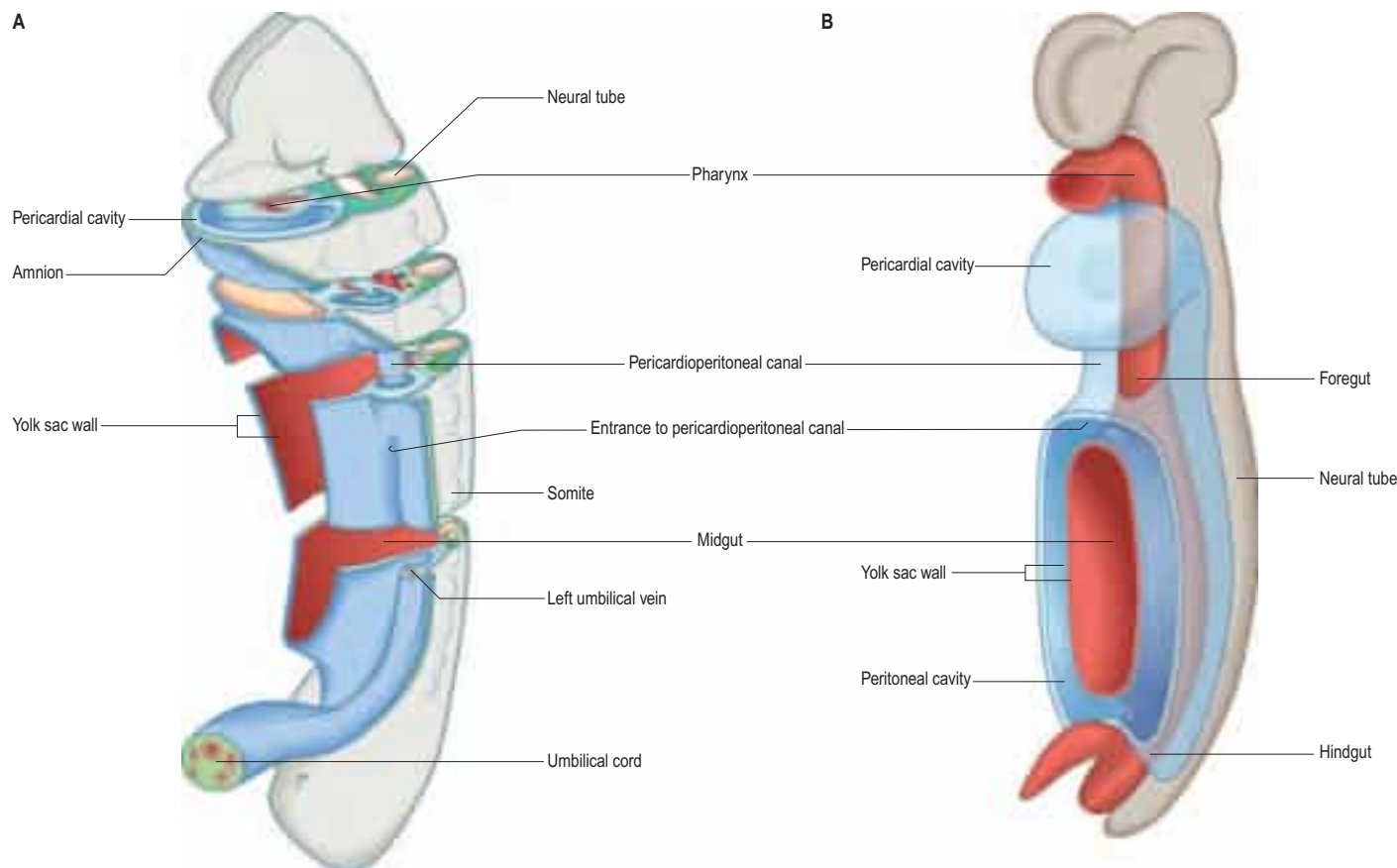


Fig. 12.1 **A**, The embryo at stage 11, showing the position of the intraembryonic coelom (contained by the walls coloured blue). **B**, The three major epithelial populations within a stage 11 embryo, viewed from a ventrolateral position. The neural tube lies dorsal to the gut. Ventrally, the intraembryonic coelom crosses the midline at the level of the foregut and hindgut, but is lateral to the midgut and a portion of the foregut.

is given below and shown in [Figures 12.1 and 12.2](#). For a summary of the fates of the embryonic cell populations, see [Figure 12.3](#).

Epithelial populations in the embryo

Surface ectoderm

During embryogenesis, the surface ectoderm shows regional differences in thickness. Ectoderm over the dorsal region of the head and trunk is thin, as is the pericardial covering; this has been interpreted as a consequence of the expansion of this epithelium over structures that are enlarging rapidly as development proceeds. After the surface ectoderm has completed a number of early interactions it forms the periderm, which remains throughout fetal life and differentiates into epidermis.

Ectodermal ring and ectodermal placodes

The ectoderm on the head and lateral borders of the embryo shows a zone of epithelial thickening, the ectodermal ring, which can be discerned from stage 10 and is completed by stage 12. Rostrally, it contains populations of neuroectoderm that remain in the ectoderm after primary neurulation and are termed ectodermal placodes; these placodes may be considered to be neuroepithelial cells that remain within the surface ectoderm until central nervous system development has progressed sufficiently for their inclusion into sensory epithelia and cranial nerve ganglia. The neuronal placodes may invaginate *in toto* to form a vesicle, or remain as a neuronal layer, or contribute individually to neuronal structures with cells of other origins. The midline ectodermal thickening, the adenohipophysial placode, invaginates as Rathke's pouch and forms a vesicle immediately rostral to the buccopharyngeal membrane. The ectodermal ring then passes bilaterally to encompass the olfactory and optic placodes, which give rise to the olfactory sensory epithelium and the lens of the eye, respectively. It then overlays the pharyngeal arches, where it gives rise to epibranchial placodes; these remove themselves individually from the ectoderm at stage 10–11 and become associated with the neural crest cells within the cranial sensory ganglia supplying the arches. It also forms specializations of the ectoderm on the frontonasal, maxillary and mandibular processes, which give rise to the tooth buds and the outer coating of the teeth. The paired otic placodes overlying the rhombencephalon at the lateral portion of the second pharyngeal arch invaginate to form the otic vesicles, which give rise to the membranous labyrinth of the ear. The ectodermal ring

then passes over the occipital and cervicothoracic parts of the embryo, superficial to the four occipital somites and later to the occipitocervical junction. Further caudally, it is associated with the upper limb field, where it will give rise to the apical ectodermal ridge. [O'Rahilly and Müller \(1985\)](#) have called the portion of the ring between the upper and lower limbs the intermembral part. It overlies the underlying intraembryonic coelom, and later (between stages 12 and 13), the mesonephric duct and ridge. In stages 14 and 15, this portion of the ectodermal ring gives rise to the mammary line. Caudal to the lower limb field, in the unfolded embryo, the ring passes distal to the cloacal membrane. In the folded embryo, this region becomes superior to the cloacal membrane and corresponds to the ectoderm associated with the external genitalia, particularly the genital tubercle and urogenital swellings, where it influences the internal and external descent of the testis and the formation of the inguinal canal.

Neural ectoderm

The neuroepithelium at the time of primary neurulation is pseudostriated. It has a midline hinge region, which, with concomitant wedging of the cells in the lateral wall of the neural groove, promotes neural tube formation. The processes of the neuroepithelium abut on to internal and external limiting membranes. This epithelium proliferates to form all the cell lines of the central nervous system and, via the production of neural crest, all the cell lines of the peripheral nervous system.

Notochord

During stage 10, the notochordal plate undergoes a process that is similar to, but a mirror image of, neurulation, and forms an epithelial tube from caudal to rostral, ending with the pharynx. The notochordal plate forms a deep groove, the vertical edges of the groove move medially and touch, and then the endodermal epithelium from each side fuses ventral to the notochord. The cells swell and develop an internal pressure (turgor) that confers rigidity on the notochord. The notochord is surrounded by a basal lamina, which is initially referred to as a perinotochordal sheath; this term is subsequently applied to mesenchymal populations that surround the notochord. After stage 11, the tubular notochord is in contact with the neural tube dorsally and the endoderm ventrally. It is not a proliferative epithelium, but it has inductive effects on the overlying neural tube and the adjacent somites, and later provides a focus for sclerotomal migration.

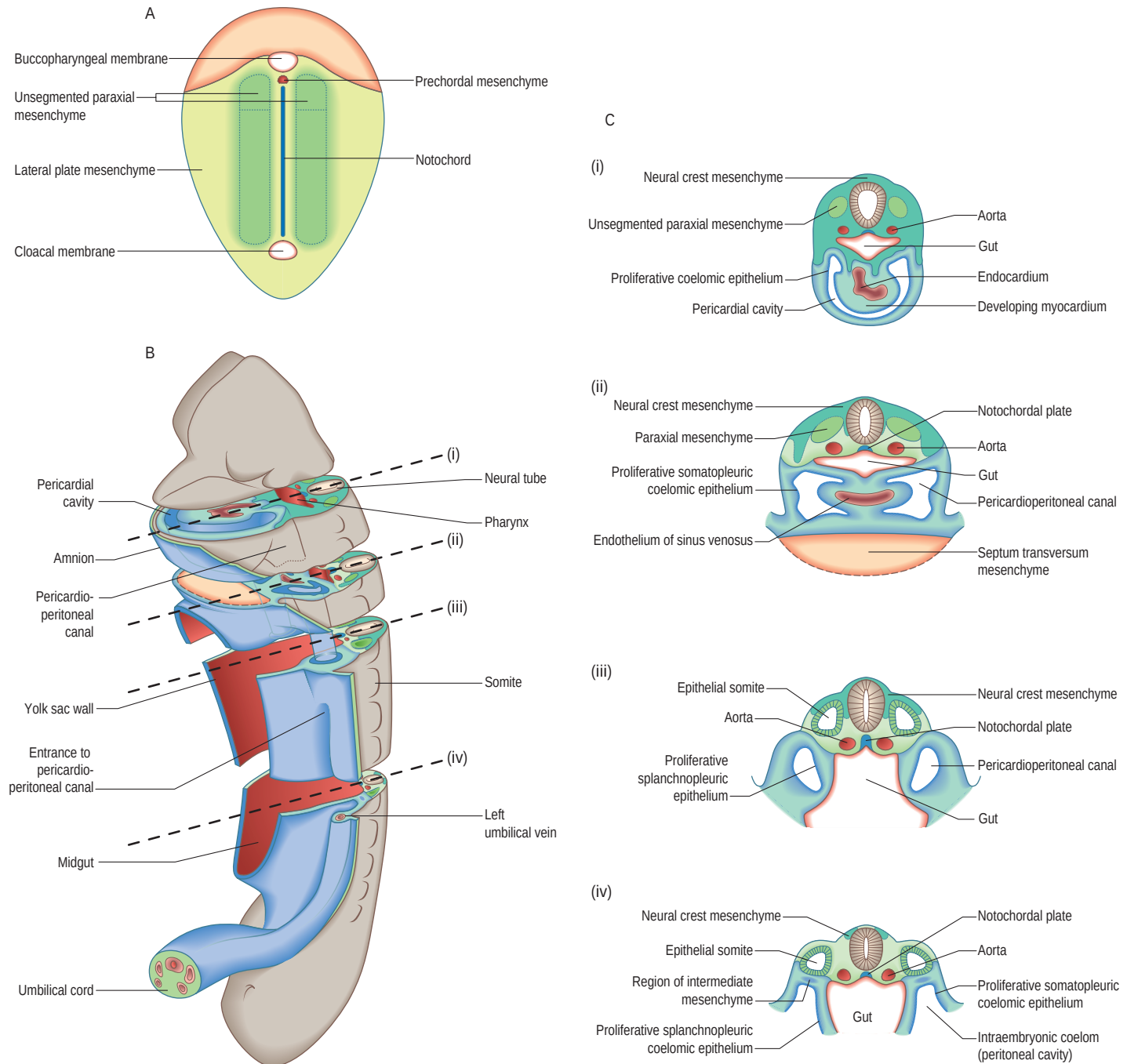


Fig. 12.2 **A**, Mesoblast populations within the early embryonic disc. **B**, A stage 11 embryo, showing the position of the intraembryonic coelom (contained by the walls coloured blue) and the positions of the sections (i)–(iv) shown in **C**. **C**, Transverse sections, arranged cranial to caudal, from a stage 11 embryo; the populations of mesenchyme and the sites of mesenchymal proliferation are indicated.

Endoderm

The craniocaudal progression of development means that the endoderm of the early stomodeum develops ahead of other portions of the endodermal epithelium. The development of the pharyngeal arches and pouches (Ch. 36) is closely associated with the development of the neural ectoderm and proliferation of the neural crest. The respiratory diverticulum arises slightly later, when the postpharyngeal gut may also be distinguished. The endoderm gives rise to the epithelial lining of the respiratory and gastrointestinal tracts, the biliary system, and the bladder and urethra.

Coelomic epithelium

The coelomic epithelium lines the intraembryonic coelom, which is subdivided into a midline pericardial cavity, two bilateral pericardioperitoneal canals, and the initially bilateral peritoneal cavities; the latter are continuous with the extraembryonic coelom. The coelomic epithelium is a germinal epithelium. It produces the myocardium and connective tissue populations for the viscera, and also gives rise to the supporting cells for the germ cells, the epithelial lining of the urogenital tracts, and the mesothelial lining of the pericardial, pleural and peritoneal cavities.

Somites

Somites are discrete epithelial spheres formed by the transformation of paraxial mesoblast cells to epithelium once their migration is complete. Specific regions of the somitic epithelium form local proliferative centres and, like neural and coelomic epithelium, it produces cells destined for different fates, at different developmental times. Somites can be seen on each side of the fusing neural tube in the human embryo from stage 9. Development proceeds in a craniocaudal direction. The original epithelial somites form the base of the skull, and the vertebral column and ribs; the dorsolateral portion, the dermomyotome, gives rise to the majority of the skeletal muscle of the body, including that in the limbs.

The relative dispositions of the neural, endodermal and coelomic epithelia are shown in Figure 12.1.

Mesenchymal populations in the embryo

In the stage 10 embryo, the major mesenchymal populations are in place. Mesoblast is still being generated at the primitive streak and moving into the presomitic mesenchymal population adjacent to the notochord. Some mesoblast is also contributing to the lateral regions

Endoderm epithelium	Coelomic wall epithelium	Mesenchyme
Primitive gut Foregut – recesses, diverticula and glands of the pharynx. General mucous glandular and duct-lining cells and the main follicular cells of the thyroid. Epithelium of pharyngeal pouches (tonsil, middle ear cavity, thymus, parathyroids 3 and 4, C cells of thyroid), adenoids, epithelial lining of the auditory tube, tympanic cavity, tympanic antrum, internal lamina of the tympanic membrane. Respiratory tract – epithelial lining, secretory and duct-lining cells of the trachea, bronchi, bronchioles and alveolar sacs. Epithelial lining, secretory and duct-lining cells of the oesophagus, stomach and duodenum. Hepatocytes of liver, biliary tract, exocrine and endocrine cells of the pancreas. Midgut – epithelial lining, glandular and duct-lining cells of the duodenum, jejunum, appendix, caecum, part of transverse colon. Hindgut – epithelial lining, glandular and duct-lining cells of part of the transverse, descending and sigmoid colon, rectum, upper part of anal canal. Allantois – urinary bladder, vagina, urethra, secretory cells of the prostate and urethral glands.	Walls of intraembryonic coelom Primitive pericardium – myocardium, parietal pericardium. Pericardioperitoneal canals – visceral, parietal and mediastinal pleura, pleuroperitoneal membranes contributing to diaphragm. <i>Splanchnopleuric epithelium</i> – visceral peritoneum of stomach, peritoneum of lesser and greater omenta, falciform ligament, lienorenal and gastrosplenic ligaments. <i>Somatopleuric epithelium</i> – parietal peritoneum. Primitive peritoneal cavity <i>Splanchnopleuric epithelium</i> – visceral peritoneal covering of mid- and hindgut, the mesentery, transverse and sigmoid mesocolon. Pronephros, epithelial lining of mesonephric ducts, vas deferens, epididymis, seminal vesicles, ejaculatory duct, ureters, vesical trigone. Müllerian ducts, epithelial lining of uterine tubes, body and cervix of uterus, vagina, broad ligament of uterus. Germinal epithelium of gonad (note the germ cells are not included on this chart because of their early sequestration into the extraembryonic tissues). Germinal epithelium forming cortex of suprarenal gland. <i>Somatopleuric epithelium</i> – parietal peritoneum, tunica vaginalis of testis.	Paraxial mesenchyme (somites and somitomeres) <i>Sclerotome</i> – vertebrae and portions of the neurocranium, axial skeleton. <i>Myotome</i> – all voluntary muscles of the head, trunk and limbs. <i>Dermatome</i> – dermis of skin over dorsal regions. Intermediate mesenchyme – connective tissue of gonads, mesonephric and metanephric nephrons, smooth muscle and connective tissues of the reproductive tracts. Septum transversum – epicardium, fibrous pericardium, portion of diaphragm, oesophageal mesentery, sinusoids of liver, tissue within lesser omentum and falciform ligament. Lateral plate mesenchyme <i>Splanchnopleuric layer</i> – smooth muscle and connective tissues of respiratory tract and associated glands. Smooth muscle and connective tissues of intestinal tract, associated glands and abdominal mesenteries. Smooth muscle and connective tissue of blood vessels (also see below). <i>Somatopleuric layer</i> – appendicular skeleton, connective tissue of limbs and trunk, including cartilage, ligaments and tendons. Dermis of ventral body wall and limbs. Mesenchyme of external genitalia. Angiogenic mesenchyme Endocardium of heart, endothelium of blood and lymphatic vessels, vessels of choroid plexus, sinusoids of liver and spleen, circulating blood cells, microglia, tissue macrophages.
Surface ectoderm epithelium	Neural plate epithelium	Neural crest
Ectodermal placodes Adenohypophysis. Sensory neurones of the cranial ganglia V, VII, VIII, IX, X. Olfactory receptor cells and olfactory epithelium. Epithelial walls of the membranous labyrinth, the cochlear organ of Corti. Lens of the eye. Enamel organs of the teeth. Cranial structures Secretory and duct-lining cells of the lacrimal, nasal, labial, palatine, oral and salivary glands. Epithelia of the cornea and conjunctiva. Epithelial lining of the external acoustic meatus and external epithelium of the tympanic membrane. Epithelial lining of the lacrimal canaliculi and nasolacrimal duct. Epithelial lining of the paranasal sinuses, lips, cheeks, gums and palate. Epidermal structures Most of the cutaneous epidermal cells, the secretory, duct-lining and myoepithelial cells of the sweat, sebaceous and mammary glands. Hair and nails. Proctodeal epithelium and epithelium of the terminal male urethra.	CNS – Brain and spinal cord Neurohypophysis. Prosencephalon (telencephalon and diencephalon) – cerebral hemispheres, basal nuclei. Mesencephalon – cerebral peduncles, tectum, tegmentum. Rhombencephalon (metencephalon and myelencephalon) – cerebellum, pons, medulla oblongata. Spinal cord. All cranial and spinal motor nerves. All CNS neurones, including preganglionic efferent neurones, with somata within the CNS. Astrocytes and oligodendrocytes. Ependyma lining the cerebral ventricles, aqueduct and central canal of brain and spinal cord, tanocytes, cells covering the choroid plexuses, circumventricular cells. Retina and optic nerve (II), epithelium of the iris, ciliary body and processes.	Neural derivatives Sensory neurones of the cranial ganglia V, VII, VIII, IX, X. Sensory neurones of the spinal dorsal root ganglia and their peripheral sensory receptors. Satellite cells in all sensory ganglia. Sympathetic ganglia and plexuses: neurones and satellite cells. Parasympathetic ganglia and plexuses: neurones and satellite cells. Enteric plexuses: neurones and glial cells. Schwann cells of all the peripheral nerves. Medulla of the suprarenal gland. Chromaffin cells. Carotid body type I cells (and type II, satellite type cells). Calcitonin-producing cells (C cells). Melanocytes. Mesenchymal derivatives in the head Frontal, parietal, squamous temporal, nasal, vomer, palatine bones, maxillae and mandible. Meninges. Choroid and sclera of eye. Connective tissue of lacrimal, nasal, labial, palatine, oral and salivary glands. Dentine of teeth. Connective tissues of head, including cartilage, ligaments and tendons. Connective tissues of thyroid gland and of the pharyngeal pouches, i.e. parathyroid glands, thymus. Tunica media of the outflow tract of the heart and the great vessels.

Fig. 12.3 Structures that will be derived from specific epithelial and mesenchymal populations in the early embryo. Abbreviation: CNS, central nervous system.

of the embryo. The different mesenchymal populations within the embryo from stage 10 onwards are described below. The relative dispositions of the early mesenchymal populations are shown in Figure 12.2.

Axial mesenchyme

The first epiblast cells to ingress through the primitive streak form the endoderm and notochord, and initially occupy a midline position. The earliest population of endodermal cells rostral to the notochordal plate is termed the prechordal plate. The notochordal cells remain medially and the endodermal cells subsequently flatten and spread laterally. The population of cells that remain mesenchymal in contact with the floor of the neural groove, just rostral to the notochordal plate, is termed prechordal mesenchyme (Fig. 12.4). These axial mesenchyme cells are tightly packed, unlike the more lateral paraxial cells but, unlike the notochord, they are not contained in an extracellular sheath. They are displaced laterally at the time of head flexion and form bilateral pre-mandibular mesenchymal condensations. They become associated with the local paraxial mesenchyme. Orthotopic grafting has demonstrated that these cells leave the edges of the prechordal mesenchyme and migrate laterally into the periocular mesenchyme, where they give rise to all of the extrinsic ocular muscles.

Paraxial mesenchyme

Paraxial mesoblast is a transient structure that forms somites at its cranial end, whilst unsegmented mesoblast is added caudally by the primitive streak. Epiblast cells that migrate through the primitive node and rostral primitive streak during gastrulation form mesoblast cells that migrate to a position lateral to the notochord and beneath the developing neural plate. Cells that ingress through the primitive node form the medial part of this paraxial mesoblast, and cells that ingress through the rostral streak form the lateral part (see Fig. 10.3). The paraxial mesoblast extends cranially from the primitive streak to the prechordal plate, which is immediately rostral to the notochord. Before somite formation, this mesenchymal tissue is also termed presomitic or unsegmented mesenchyme in mammals (analogous to the segmental plate in birds). Paraxial mesoblast rostral to the otic vesicle was previously believed not to segment.

Caudal to the otic vesicle, the paraxial mesoblast on each side of the rhombencephalon segments into somites as the neural folds elevate and neurulation begins; somites are therefore post-otic. During somitogenesis, mesoblastic cells show changes in shape and in cell–cell adhesion, and become organized into epithelial somites; this process begins at the eighth somitomere, just caudal to the midpoint of the notochordal plate. Somite one is also termed the first occipital somite. The post otic paraxial cell population is termed paraxial mesoderm in contemporary literature.

Skeletal muscle throughout the body is derived from paraxial mesoblast-derived somatic epithelium, which proliferates to form myoblastic populations that migrate to the head, body and limbs.

Septum transversum

Early mesenchyme that invaginates through the middle part of the primitive streak comes to lie rostral to the buccopharyngeal membrane, where the cells form the epithelial wall of the pericardial coelom. As this epithelium proliferates, the visceral pericardial wall gives rise to myocardium. The parietal pericardial wall forms mesenchyme, initially termed precardiac, or cardiac, mesenchyme, which is able to induce proliferation of hepatic endodermal epithelium. With further proliferation, the precardiac mesenchyme forms a ventral mass caudal to the heart, the septum transversum, separating the foregut endoderm from the pericardial coelom (see Fig. 12.2). By stage 11, the septum transversum extends dorsally on each side of the developing gut and becomes continuous with the mesenchymal populations that proliferate from the walls of the pericardioperitoneal canals. Cells of the septum transversum give rise to the sinusoids of the liver, the central portions of the diaphragm and the epicardium.

Neural crest

The neural crest is unique: it gives rise to neural populations in the head and trunk, and also provides an extensive mesenchymal population in the head with attributes similar, in terms of patterning, to somatopleuric mesenchyme. Neural crest cells arise from cells that lie initially at the outermost edges of the neural plate, between the presumptive epidermis and the neural tube, and are committed to a neural crest lineage before the neural plate begins to fold. After neurulation, neural crest cells form a transient axial population and then disperse, in some cases migrating over considerable distances, to a variety of different developmental fates. Unlike mesoblast, which is produced from

the primitive streak, none of the cells that arise from the neural crest become arranged as epithelia. The development and fate of head and trunk neural crest cells are very different and therefore they will be considered separately.

Head neural crest

Head neural crest migrates before the neural tube closes. Two populations of crest cells develop. Some retain a neuronal lineage and contribute to the somatic sensory and parasympathetic ganglia in the head and neck. Others produce extensive mesenchymal populations; the crest cell population arising from the head is larger than that found at any trunk level. Each brain region has its own crest population that migrates dorsolaterally around the sides of the neural tube to reach the ventral side of the head. Crest cells surround the prosencephalic and optic vesicles, and occupy each of the pharyngeal arches (Ch. 36). They provide mesenchyme cells that will produce the connective tissue in parts of the neuro- and viscerocrania. All cartilage, bone, ligament, tendon, dermal components and glandular stroma in the head are derived from the head neural crest. Head neural crest also contributes to the tunica media of the aortic arch arteries.

Trunk neural crest

Unlike its counterpart in the head, trunk neural crest is formed as the neural tube closes craniocaudally, which means that various stages of crest development can be found in the more caudal regions of an embryo. As the neural tube begins to fuse dorsally in the midline, the neural crest cells lose their epithelial characteristics and junctional connections, and form a band of loosely arranged mesenchyme cells immediately dorsal to the neural tube and beneath the ectoderm. Initially, most of the crest cells lie with their long axes perpendicular to the long axis of the neural tube. Later, the cell population expands laterally and around the neural tube as a sheet. Trunk neural crest cells move from their position dorsal to the neural tube via three routes (see Fig. 17.11): dorsolaterally, to form dorsal root ganglia throughout the trunk; ventrally, to form sympathetic ganglia, enteric nerves and the suprarenal medulla; and rostrocaudally, along the aorta, to form the pre-aortic ganglia. In a second migration route, crest cells pass dorsolaterally between the ectoderm and the epithelial plate of the somite into the somatopleure, where they eventually form melanocytes in the skin.

Lateral plate

Lateral plate is the term for the early unsegmented mesoblast population lateral to the paraxial mesenchyme. Mesoblastic cells, which arise from the middle of the primitive streak (primary mesenchyme), migrate cranially, laterally and caudally to reach their destinations, where they revert to epithelium and form a continuous layer that adheres to the ectoderm dorsally and the endoderm ventrally. The epithelium faces a new intraembryonic cavity, the intraembryonic coelom, which becomes confluent with the extraembryonic coelom and provides a route for the circulation of coelomic fluid through the embryo. Once formed, the intraembryonic coelomic wall becomes a proliferative epithelium that produces new populations of mesenchymal cells. The mesenchymal population subjacent to the ectoderm is termed somatopleuric mesenchyme and is produced by the somatopleuric coelomic epithelium. The mesenchymal population surrounding the endoderm is termed splanchnopleuric mesenchyme and is produced by the splanchnopleuric coelomic epithelium (see Fig. 12.2).

It is important to note that these terms are relevant only caudal to the third pharyngeal arch. Rostral to this, there is a sparse mesenchymal population between the pharynx and the surface ectoderm prior to migration of the head neural crest, and there are no landmarks with which to demarcate lateral from paraxial mesenchyme. This unsplit lateral plate is believed to contribute to the cricoid and arytenoid cartilages, the tracheal rings and the associated connective tissue.

Somatopleuric mesenchyme

Somatopleuric mesenchyme produces a mixed population of connective tissues and has a significant organizing effect at the level of the developing limbs. The pattern of limb development is controlled by information contained in the somatopleuric mesenchyme. Regions of the limb are specified by interaction between the surface ectoderm (apical ectodermal ridge) and underlying somatopleuric mesenchyme; together, these tissues form the progress zone of the limb. The somatopleuric mesenchyme in the limb bud also specifies the postaxial border of the developing limb. Somatopleuric mesenchyme gives rise to the connective tissue elements of the appendicular skeleton, including the pectoral and pelvic girdles and the bones and cartilage of the limbs, and their associated ligaments and tendons. It also gives rise to the dermis of the skin of the ventral and lateral body walls and of the limbs.

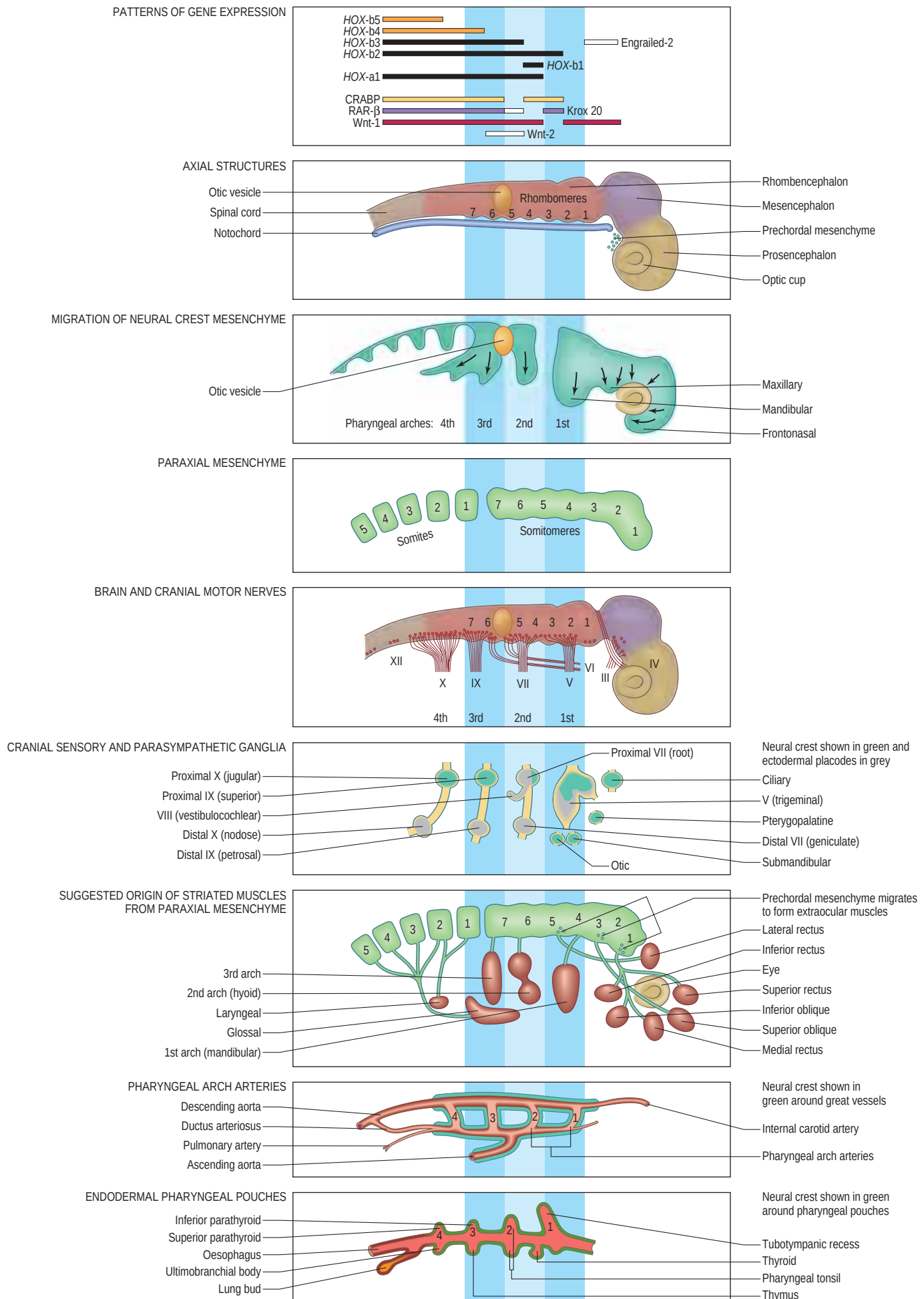


Fig. 12.4 The organization of the head and pharynx in an embryo at about stage 14. The individual tissue components have been separated but are aligned in register through the numbered zones.

Splanchnopleuric mesenchyme

Splanchnopleuric mesenchyme surrounds the developing gut and respiratory tubes, contributing connective tissue cells to the lamina propria and submucosa, and smooth muscle cells to the muscularis mucosae and muscularis externa. It plays a patterning role in endodermal development, specifying the region and villus type in the gut, and the branching pattern in the respiratory tract.

Intermediate mesenchyme

Intermediate mesenchyme is a loose collection of cells found between the somites and the lateral plate (see Fig. 12.2). Its development is closely related to the progress of differentiation of the somites and the proliferating coelomic epithelium from which it is derived. Intermediate mesenchyme is not present before somitogenesis or the formation of the eighth somite. In embryos with 8–10 somites, it is present lateral to the sixth somite but does not extend cranially. The mesenchyme cells are arranged as layers, one continuous with the dorsal side of the paraxial mesenchyme and the somatopleure, the other with the ventral side of the paraxial mesenchyme and the splanchnopleure.

As development proceeds, the intermediate mesenchyme forms a loosely packed dorsolateral cord of cells, which lengthens at the caudal end and ultimately joins the cloaca. It gives rise to the nephric system, gonads and reproductive ducts.

Angioblastic mesenchyme

Mesenchymal cells, which give rise to the cellular elements of the blood, the endothelium and the mesenchymal layers of the tunica externa and adventitia of blood and lymphatic vessels, arise from extraembryonic and intraembryonic sources. Evidence suggests that

endodermal tissues are necessary for endothelial differentiation, particularly the early foregut. Angioblastic mesenchyme forms early in the third week of development from extraembryonic mesenchyme in the splanchnopleure of the yolk sac, in the body stalk (containing the allantois), and in the somatopleure of the chorion. The peripheral cells flatten as a vascular endothelium, whereas the central cells transform into primitive red blood corpuscles. Later, contiguous islands merge, forming a continuous network of fine vessels. Intraembryonic blood vessels are first seen at the endoderm–mesenchyme interface within the lateral splanchnic mesenchyme at the caudolateral margins of the cranial intestinal portal. Angioblastic competence has been demonstrated within the ventral (splanchnopleuric) mesenchymes with which the endoderm interacts. However, the notochord and prechordal plate do not contain angiogenic cells. Similarly, ectodermal tissues do not appear to give rise to angiogenic cells. Somites, derived from paraxial mesenchyme, have been shown to be a source of angioblasts that either differentiate with the somite derivatives, or migrate to the neural tube, ventrolateral body wall, limb buds, mesonephros and the dorsal part of the aorta.

The earliest angiogenic mesenchymal cells form blood vessels by vasculogenesis, a process in which new vessels (e.g. endothelial heart tubes, dorsal aortae, umbilical and early vitelline vessels) develop *in situ* (Ch. 13). Later vessels develop by angiogenesis, sprouting and branching from the endothelium of pre-existing vessels; this process is the means by which most other vessels develop. The ultimate pattern of vessels is controlled by the surrounding, non-angiogenic mesenchyme; vessels become morphologically specific for the organ in which they develop, and also immunologically specific, expressing organ-specific proteins.

KEY REFERENCES

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